

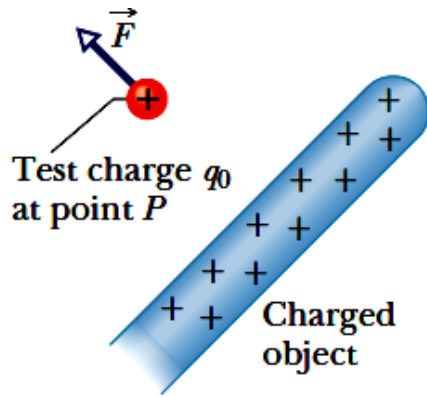
Halliday & Resnick's
Fundamentals of Physics
Extended Edition

Jearl Walker

Chapter 22

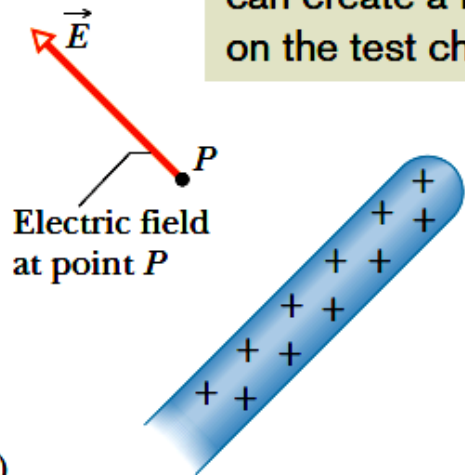
Electric Fields

22.2 The Electric Field:



(a)

The rod sets up an electric field, which can create a force on the test charge.



(b)

Fig. 22-1 (a) A positive test charge q_0 placed at point P near a charged object. An electrostatic force $\vec{F}$ acts on the test charge. (b) The electric field $\vec{E}$ at point P produced by the charged object.

The Electric Field is a *vector field*.

The electric field, $\vec{E}$, consists of a distribution of vectors, one for each point in the region around a charged object, such as a charged rod.

We can define the electric field at some point near the charged object, such as point P in Fig. 22-1a, as follows:

- A positive test charge q_0 , placed at the point will experience an electrostatic force, $\vec{F}$.
- The electric field at point P due to the charged object is defined as the electric field, $\vec{E}$, at that point:

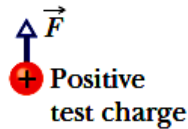
$$\vec{E} = \frac{\vec{F}}{q_0} \quad (\text{electric field}).$$

The SI unit for the electric field is the newton per coulomb (N/C).

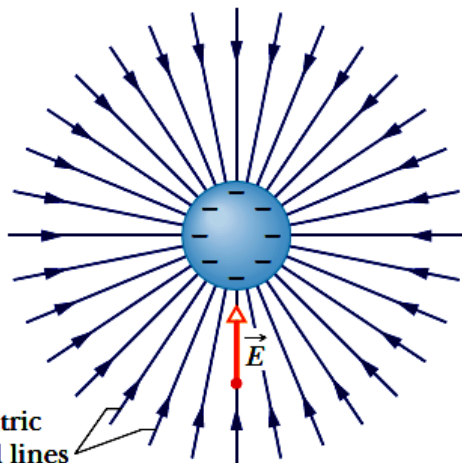
22.2 The Electric Field:

Table 22-1	
Some Electric Fields	
Field Location or Situation	Value (N/C)
At the surface of a uranium nucleus	3×10^{21}
Within a hydrogen atom, at a radius of 5.29×10^{-11} m	5×10^{11}
Electric breakdown occurs in air	3×10^6
Near the charged drum of a photocopier	10^5
Near a charged comb	10^3
In the lower atmosphere	10^2
Inside the copper wire of household circuits	10^{-2}

22.3 Electric Field Lines:



(a)



(b)

Electric field lines

Electric field lines extend away from positive charge (where they originate) and toward negative charge (where they terminate).

- At any point, the direction of a straight field line or the direction of the tangent to a curved field line gives the direction of $\vec{E}$ at that point.
- The field lines are drawn so that the number of lines per unit area, measured in a plane that is perpendicular to the lines, is proportional to the magnitude of $\vec{E}$.

Thus, $\vec{E}$ is large where field lines are close together and small where they are far apart.

Fig. 22-2 (a) The electrostatic force $\vec{F}$ acting on a positive test charge near a sphere of uniform negative charge. (b) The electric field vector $\vec{E}$ at the location of the test charge, and the electric field lines in the space near the sphere. The field lines extend *toward* the negatively charged sphere. (They originate on distant positive charges.)

22.3 Electric Field Lines:

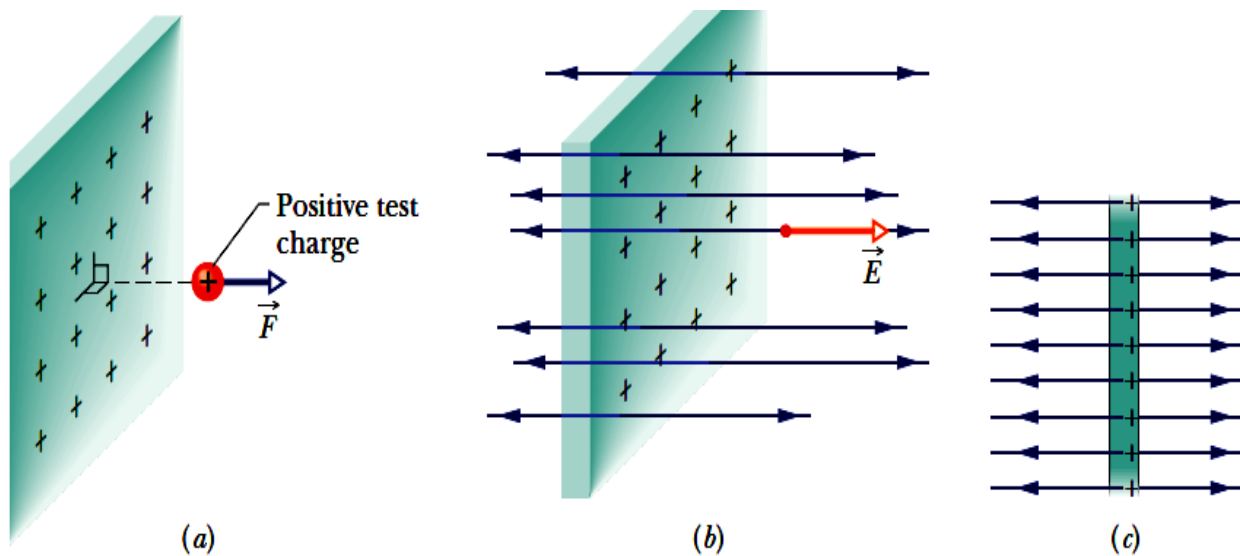
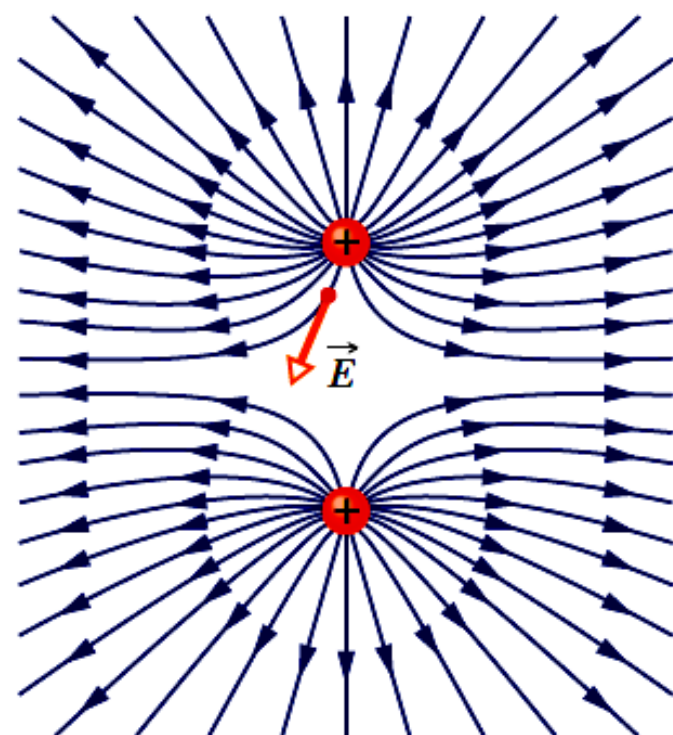


Fig. 22-3 (a) The electrostatic force $\vec{F}$ on a positive test charge near a very large, nonconducting sheet with uniformly distributed positive charge on one side. (b) The electric field vector $\vec{E}$ at the location of the test charge, and the electric field lines in the space near the sheet. The field lines extend *away from* the positively charged sheet. (c) Side view of (b).

Fig. 22-4 Field lines for two equal positive point charges. The charges repel each other. (The lines terminate on distant negative charges.) The electric field vector at one point is shown; note that it is tangent to the field line through that point.



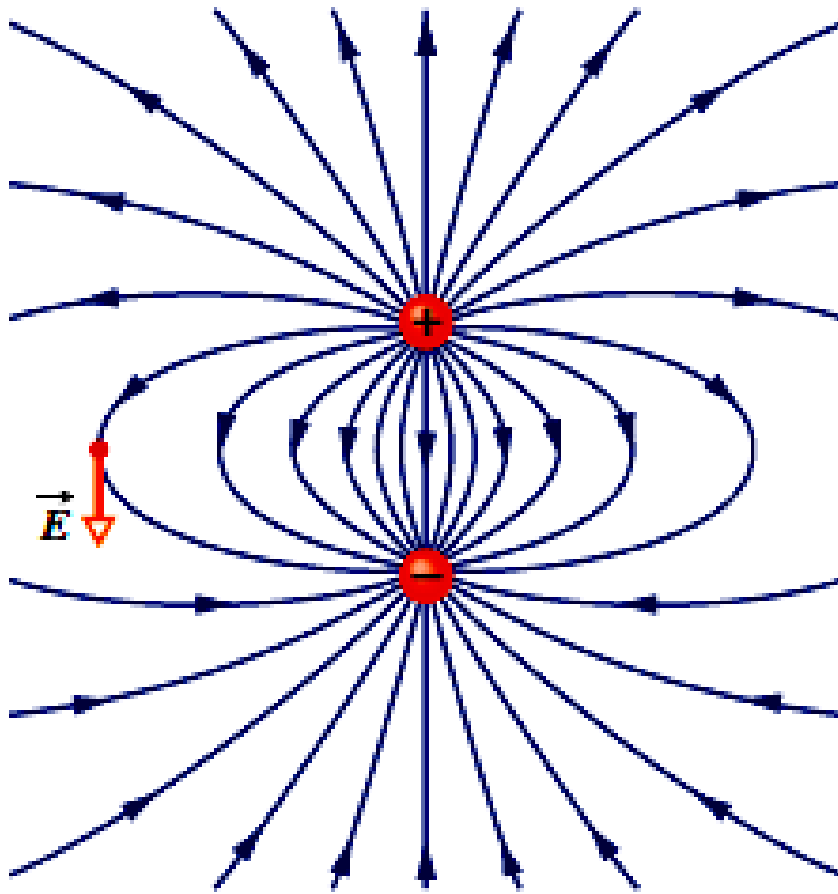


Fig. 22-5 Field lines for a positive point charge and a nearby negative point charge that are equal in magnitude. The charges attract each other. The pattern of field lines and the electric field it represents have rotational symmetry about an axis passing through both charges in the plane of the page. The electric field vector at one point is shown; the vector is tangent to the field line through the point.

22.4 The Electric Field due to a Point:

To find the electric field due to a point charge q (or charged particle) at any point a distance r from the point charge, we put a positive test charge q_0 at that point.

$q_1=q; q_2=q_0$

$$\vec{E} = \frac{\vec{F}}{q_0} \quad (\text{electric field}).$$

The direction of E is directly away from the point charge if q is positive, and directly toward the point charge if q is negative. The electric field vector is:

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r} \quad (\text{Coulomb's law}),$$

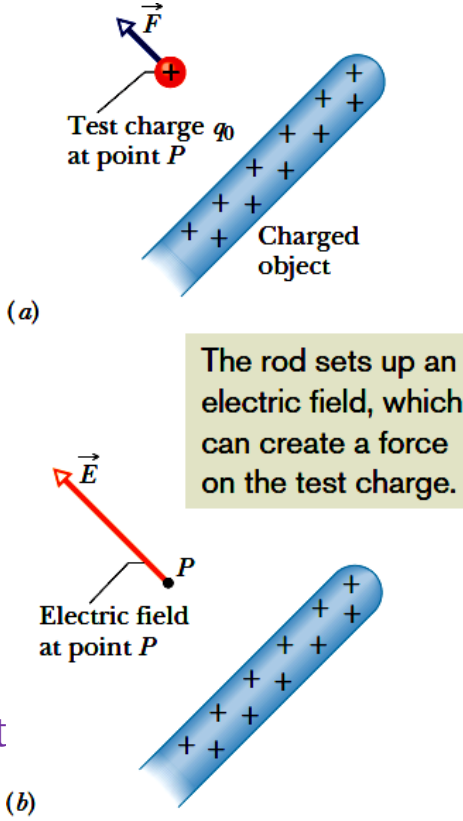
$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{point charge}).$$

The net/resultant, electric field due to more than one point charge can be found by the superposition principle. If we place a positive test charge q_0 near n point charges $q_1, q_2, \dots, q_n$, then, the net force, $\vec{F}_0$, from the n point charges acting on the test charge is

$$\vec{F}_0 = \vec{F}_{01} + \vec{F}_{02} + \dots + \vec{F}_{0n}.$$

The net electric field at the position of the test charge is

$$\begin{aligned} \vec{E} &= \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \dots + \frac{\vec{F}_{0n}}{q_0} \\ &= \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n. \end{aligned}$$



22.5 The Electric Field due to an Electric Dipole:

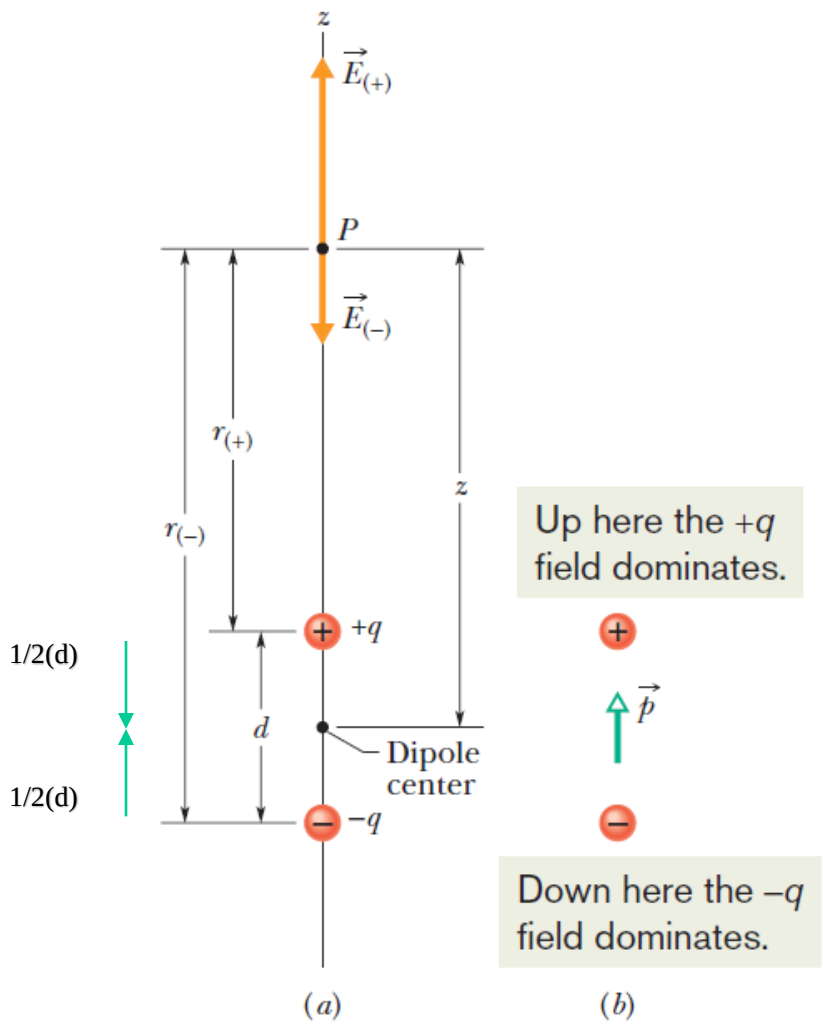
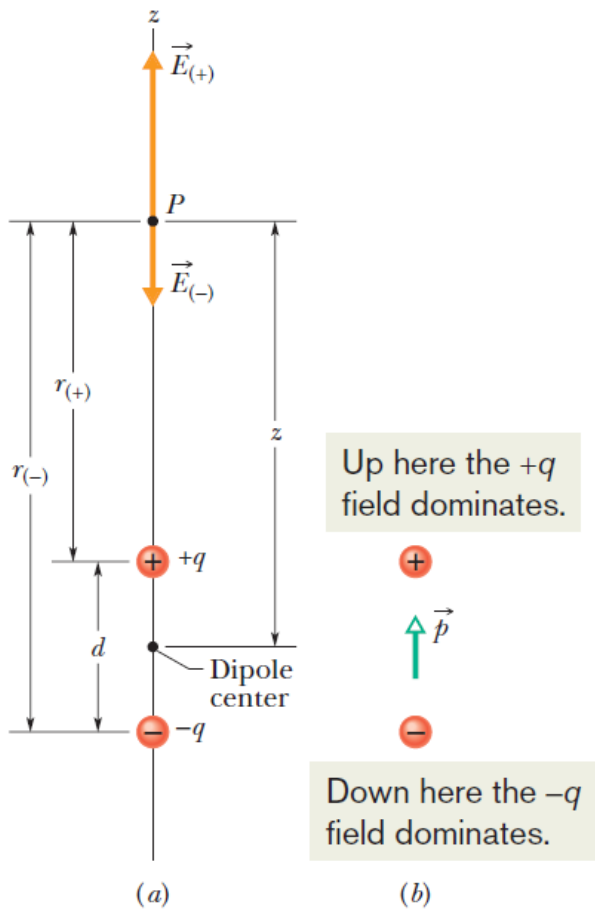


Fig. 22-8 (a) An electric dipole. The electric field vectors $\vec{E}_{(+)}$ and $\vec{E}_{(-)}$ at point P on the dipole axis result from the dipole's two charges. Point P is at distances $r_{(+)}$ and $r_{(-)}$ from the individual charges that make up the dipole. (b) The dipole moment $\vec{p}$ of the dipole points from the negative charge to the positive charge.

$$r_{(+)} = z - 1/2(d)$$
$$r_{(-)} = z + 1/2(d)$$

22.5 The Electric Field due to an Electric Dipole:

From symmetry, the electric field E at point P —and also the fields E_+ and E_- due to the separate charges that make up the dipole—must lie along the dipole axis, which we have taken to be a z axis. From the superposition principle for electric fields, the magnitude E of the electric field at P is



$$\begin{aligned}
 E &= E_{(+)} - E_{(-)} \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(-)}^2} \\
 &= \frac{q}{4\pi\epsilon_0(z - \frac{1}{2}d)^2} - \frac{q}{4\pi\epsilon_0(z + \frac{1}{2}d)^2}.
 \end{aligned}$$

$$\begin{aligned}
 E &= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right) \\
 E &= \frac{q}{4\pi\epsilon_0 z^2} \frac{2d/z}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2} = \frac{q}{2\pi\epsilon_0 z^3} \frac{d}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2} \\
 d/2z \ll 1 &\longrightarrow E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}.
 \end{aligned}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3} \quad (\text{electric dipole}).$$

The product qd , which involves the two intrinsic properties q and d of the dipole, is the magnitude p of a vector quantity known as the **electric dipole moment** of the dipole.

(a) Electric dipole - The electric fields $\vec{E}_+$ & $\vec{E}_-$ at point P on the dipole axis result from the dipole's two charges.

$$\vec{E} = E_{(+)} + (-E_{(-)})$$

$$\vec{E} = \vec{E}_{(+)} - \vec{E}_{(-)}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{(z - \frac{d}{2})^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(z + \frac{d}{2})^2} \rightarrow (22.5)$$

$$r(+) = z - 1/2(d)$$

$$r(-) = z + 1/2(d)$$

After little algebra, the working eq.

$$\vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(z - \frac{d}{2})^2} - \frac{1}{(z + \frac{d}{2})^2} \right]$$

x 2 ÷ by z^2 on RHS of eq.

$$\vec{E}_{\text{net}} = \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{z^2}{(z - \frac{d}{2})^2} - \frac{z^2}{(z + \frac{d}{2})^2} \right]$$

$$\Rightarrow z = \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{1}{(\frac{z}{z} - \frac{d}{2z})^2} - \frac{1}{(\frac{z}{z} + \frac{d}{2z})^2} \right]$$

$$E_{\text{net}} = \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{1}{(1 - \frac{d}{2z})^2} - \frac{1}{(1 + \frac{d}{2z})^2} \right]$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left[\left(1 - \frac{d}{2z}\right)^{-2} - \left(1 + \frac{d}{2z}\right)^{-2} \right]$$

Binomial series -

$$(1+x)^{\alpha} = 1 + \frac{\alpha x}{1!} + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$$

$$\text{or } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$$

$$\text{or } (1+x)^{-2} = 1 + (-2)x + \frac{(-2)(-2-1)}{2!} x^2 + \dots$$

$$\alpha = -2, x = \frac{d}{2z}$$

$$\left(1 - \frac{d}{2z}\right)^{-2} = 1 + (-2) \cdot \frac{d}{2z} + \frac{(-2)(-2-1)}{2!} \left(\frac{d}{2z}\right)^2 + \dots$$

ignoring higher order terms.

$$= \left(1 + \frac{d}{z}\right)$$

$$\left(1 + \frac{d}{2z}\right)^{-2} = 1 + (-2) \cdot \frac{d}{2z} + \frac{(-2)(-2-1)}{2!} \left(\frac{d}{2z}\right)^2 + \dots$$

$$= \left(1 - \frac{d}{z}\right)$$

$$E_{\text{net}} = \frac{q}{4\pi\epsilon_0 z^2} \left[\left(1 + \frac{d}{z}\right) - \left(1 - \frac{d}{z}\right) \right]$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left[1 + \frac{d}{z} - 1 + \frac{d}{z} \right]$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{d+d}{z} \right)$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{2d}{z} \right)$$

$$\boxed{\vec{F}_{\text{net}} = \frac{qd}{2\pi\epsilon_0 z^3}}$$

Electric field due to
electric dipole
direction is from
-ve to +ve.
Electric $\vec{p}$ (Cm)
dipole
moment

Example, Electric Dipole and Atmospheric Sprites:

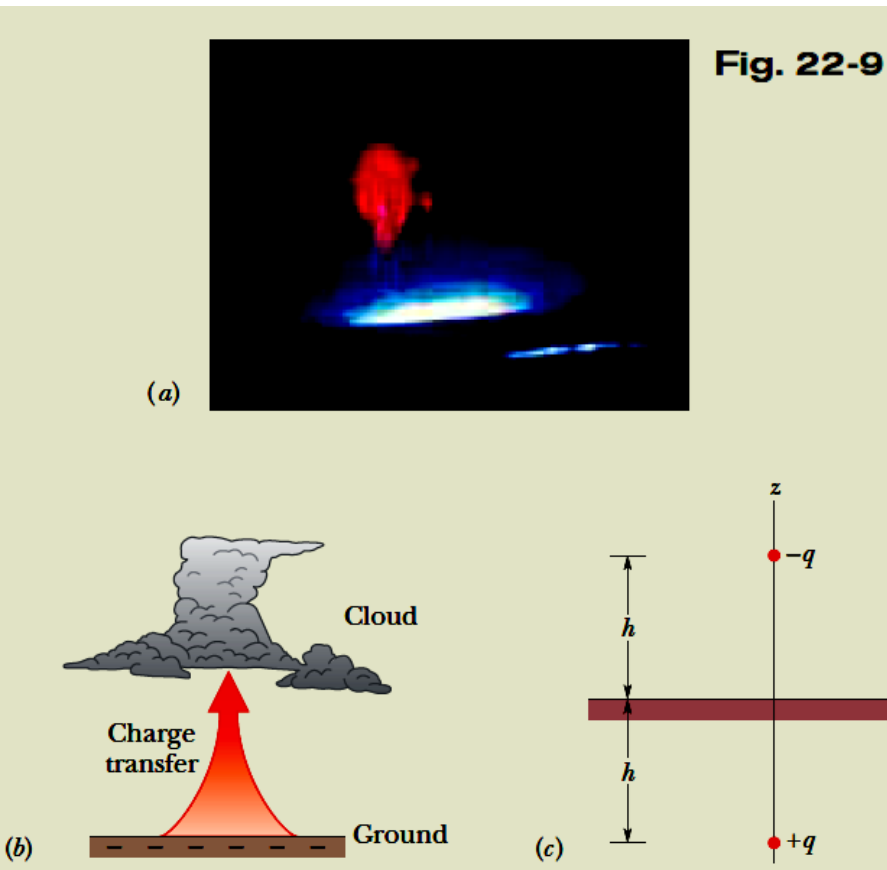


Fig. 22-9

We can model the electric field due to the charges in the clouds and the ground by assuming a vertical electric dipole that has charge $-q$ at cloud height h and charge $+q$ at below-ground depth h (Fig. 22-9c).

If $q = 200 \text{ C}$ and $h = 6.0 \text{ km}$, what is the magnitude of the dipole's electric field at altitude $z_1 = 30 \text{ km}$ somewhat above the clouds and altitude $z_2 = 60 \text{ km}$ somewhat above the stratosphere?

$$E = \frac{1}{2\pi\epsilon_0} \frac{q(2h)}{z^3},$$

where $2h$ is the separation between $-q$ and $+q$ in Fig. 22-9c. For the electric field at altitude $z_1 = 30 \text{ km}$, we find

$$\begin{aligned} E &= \frac{1}{2\pi\epsilon_0} \frac{(200 \text{ C})(2)(6.0 \times 10^3 \text{ m})}{(30 \times 10^3 \text{ m})^3} \\ &= 1.6 \times 10^3 \text{ N/C.} \end{aligned} \quad (\text{Answer})$$

Similarly, for altitude $z_2 = 60 \text{ km}$, we find

$$E = 2.0 \times 10^2 \text{ N/C.} \quad (\text{Answer})$$

Sprites (Fig. 22-9a) are huge flashes that occur far above a large thunderstorm. They are still not well understood but are believed to be produced when especially powerful lightning occurs between the ground and storm clouds, particularly when the lightning transfers a huge amount of negative charge $-q$ from the ground to the base of the clouds (Fig. 22-9b).

22.6 The Electric Field due to a Continuous Charge:

When we deal with continuous charge distributions, it is most convenient to express the charge on an object as a *charge density rather than as a total charge*. For a line of charge, for example, we would report the *linear charge density* (or charge per unit length) λ , whose SI unit is the coulomb per meter.

Table 22-2 shows the other charge densities we shall be using.

Table 22-2		
Some Measures of Electric Charge		
Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m ²
Volume charge density	ρ	C/m ³

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Some Measures of Electric Charge

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m ²
Volume charge density	ρ	C/m ³

charge per unit length

charge per unit area

charge per unit volume

$$\lambda = \frac{q}{L} \text{ C/m}$$

$$\sigma = \frac{q}{A} \text{ C/m}^2$$

$$\rho = \frac{q}{\text{Vol.}} \text{ C/m}^3$$

22.8: A Point Charge in an Electric Field

The electrostatic force $\vec{F}$ acting on a charged particle located in an external electric field $\vec{E}$ has the direction of $\vec{E}$ if the charge q of the particle is positive and has the opposite direction if q is negative.

$$\vec{F} = q\vec{E},$$

When a charged particle, of charge q , is in an electric field, $\vec{E}$, set up by other stationary or slowly moving charges, an electrostatic force, $\vec{F}$, acts on the charged particle as given by the above equation.

Electrical Breakdown and Sparking

If the magnitude of an **electric field in air exceeds a certain critical value E_c** , the air undergoes electrical breakdown, a process whereby the **field removes electrons from the atoms in the air**.

The air then begins to **conduct electric current** because the freed electrons are propelled into motion by the field.

As they move, they collide with any atoms in their path, causing those **atoms to emit light**.

We can see the paths, commonly called **sparks**, taken by the freed electrons because of that emitted light.

Figure 22-18 **shows sparks above charged metal wires** where the electric fields due to the wires cause electrical breakdown of the air.

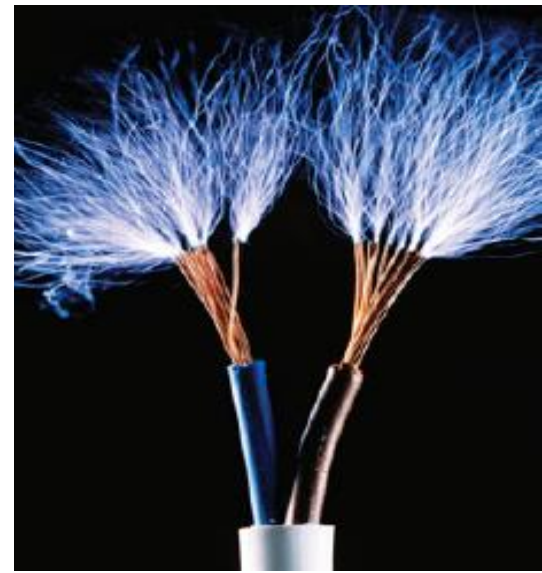


Figure 22-18 The metal wires are so charged that the electric fields they produce in the surrounding space cause the air there to undergo electrical breakdown.